

Team Project

Purpose: Analysis, design, and implementation of a relational database using Universal Relation Approach.

Problem: The publishing company of Focus published the book of “Glorious Savannah” which includes information about every aspect of savannah. The book has four major sections of: Attractions, Businesses, Hotels, and Restaurants. Each section may have several sub-sections. The authorities of the Focus Company want you to analyze, design, and implement a database for the attached portion of their book. The analysis step produces the set of semantic rules regarding the book. The design step produces a set of relations that are all in 1NF, 2NF, 3NF, BCNF, and 4NF. The implementation step uses a relational database management system (except Access DBMS) to actually create, populate, and manipulate the database. To manipulate the database:

(A) Create two views as follow:

View1: Get the name of family owned and operated businesses that have the same area code as the businesses with offer of free estimates.

View2: List of historical churches that are located on one of the Savannah’s squares.

(B) Issue the following queries:

1. Get the list of businesses that have an 800 number, a website, and accept the credit cards of Visa, Master Card, Discover, and American Express.
2. Get all the information about all the steakhouses in Hilton Head
3. Get the name of businesses that are not located in Savannah.
4. Get the name of pest controllers whose expertise is in dead animal removal from both residential and commercial areas.
5. Get the name of optical doctors of Georgia Eye Institute of the Southeast.
6. Get the name of plastic surgery establishment who offer eye lid surgery.
7. Get the list of art activities in savannah that includes Jazz music.
8. Get the list of public golf courses in savannah with more than 20-holes.
9. Get the address and name of sports recreation for Tybee Island.
10. Get the name of squares in Savannah established for war heroes.
11. Get the name and address of public parks that includes both of picnic shelters and one form of athletic field.
12. Get the list of all the Art events that take place in Savannah during the months of March and April.
13. Get the list of Savannah’s historic area hotels that charge a fee for cancellation of a reservation and they have a limitation on the number of days that one can book a room.

14. Get the list of those businesses that have been in business for more than 10 years, willing to accept Master Card, and they are family owned and operated.
15. The Southcoast Medical group has closed its primary care practice in Richmond Hill and it also closed all of its cardiology practices. However the medical group added the cardiology practice in Garden City. Update your database to reflect the new changes.
16. Get the name of all the savannah attractions that have been erected in 19th century.
17. Insert the following square in the database:
AASU Square named for Michael Jordan who served as president of AASU between 1994-2000. The square is located at the Apache Street and Abercorn and it was laid out in 2007 by James Cameron using the Hungarian architecture.

(C) Your database must not be designed to support only the above queries. You will be asked to demonstrate in class the performance of a new query that is not in the above list but it is related to the content of the book. This demonstration is a part of the grade for the course.

Turn in a highly organized report (One per team) that includes the analysis, design, and implementation of the database and queries. The report must have three distinct parts separated by colored dividers and one appendix. The first part includes the analysis and design of the database. The second part includes Implementation of the database (*creating* database, *populating* database, and *displaying* the content of all database relations.) Part three includes only the implementation of the above queries and for each query the following format must be followed:

Query/View number:

Query text

SQL codes

The query output generated by MYSQL

*// I am interested to see the actual output of MYSQL and not a cleaned version of it. The cleaned
// version receives score of zero.*

The appendix content will be described shortly.

Each team presents and defends its analysis, design, and implementation of the project in the classroom on the *presentation day* (date of the presentation day will be announced in the classroom.) Presentation is formal (tie and jacket is required.) It is crucial that every team member be equally prepared to act as a presenter and defender of the project because the presenter will be chosen at the time of presentation by the professor and not by the team. The presenter will be seriously challenged about every aspects of the project during the presentation. At the time of presentation only one team will be in the classroom. All presenters must use power point slides and be able to talk about the content of slides and not reading the slides' contents from the screen or from the notes.

In addition, two new queries will be given to each team on the *demonstration day* (date of the demonstration day will be announced in the classroom.) The team members must write SQL

codes for the given queries, run the codes against their database, and show the results in an hour time. Make sure that you bring with you the computer that your database is resided on the demonstration day.

NOTES:

Team members are chosen randomly as is the case in real life (you do not pick your co-workers.) One team member is designated as the captain of the team. The duties of the captain are:

1. To keep minutes of every meeting by filling the following form for each meeting. **The minutes of meetings make the appendix of the final report.**

MINUTES OF THE MEETING		
Date of Meeting:	Start time:	End time:
Name of members who were present:		
Name of members who were late (The amount of time that member was late):		
Members' duty assignments, and progress in reference to the previous meeting:		
Name:	Duty:	Progress:
Name:	Duty:	Progress:
Name:	Duty:	Progress:
Name:	Duty:	Progress:
Name:	Duty:	Progress:
Time set for the next meeting:		
Signature of all the present members:		
Member 1	Member2	
Member 3	Member4	
Member 5		

2. To prepare the *Contribution Sheet* that includes names of team members along with their overall percentage of contribution to the assignment and their signatures. The Contribution Sheet goes after the cover letter in the final report. Any dispute on the percentage of contribution will be finalized by me using the minutes of the meetings.
3. If a member of the team does not participate regularly in the meetings or does not finish his/her assigned task, then team (not the Capitan) members must collectively make an appointment with me and report the lack of participation by the team member as soon as possible. The undedicated member will be separated from the team immediately and works on the assignment as a team of one. The same final report is expected from the team of one.
4. If a team has a question about the assignment, then the entire team must be in my office when the question is asked. This means, I will not respond to a question about any aspects of the assignment if it is asked by one member of the team. In the case of emergency, the presence of the majority of the team members is required.